

Tugas 2: Laporan Praktikum Mandiri 2

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```
from google.colab import drive
drive.mount('/content/gdrive')

Mounted at /content/gdrive

path = "/content/gdrive/MyDrive/machine learning/praktikum/praktikum2"

import pandas as pd

df = pd.read_csv(path + '/data/500_Person_Gender_Height_Weight_Index.csv')
df
```

	Gender	Height	Weight	Index
0	Male	174	96	4
1	Male	189	87	2
2	Female	185	110	4
3	Female	195	104	3
4	Male	149	61	3
...	...	...	...	...
495	Female	150	153	5
496	Female	184	121	4
497	Female	141	136	5
498	Male	150	95	5
499	Male	173	131	5

500 rows x 4 columns

Cell1: Menghubungkan google colab ke google drive agar file di drive bisa diakses

Cell2: Menyimpan lokasi folder kerja (path) tempat file dataset berada

Cell3: Mengimpor library pandas, membaca file CSV ke dataframe df, dan menampilkan seluruh data

```
import pandas as pd
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 4 columns):
#   Column   Non-Null Count  Dtype  
---  -
0    Gender   500 non-null    object  
1    Height   500 non-null    int64   
2    Weight   500 non-null    int64   
3    Index     500 non-null    int64   
dtypes: int64(3), object(1)
memory usage: 15.8+ KB

df['Height'].mean()

np.float64(169.944)

df['Height'].median()

170.5

df['Height'].mode()

Height
0      188
dtype: int64
```

Cell4: Menampilkan informasi umum dataframe seperti jumlah baris, kolom, dan tipe data

Cell5: Menghitung nilai rata-rata (mean) dari kolom height

Cell6: Menghitung nilai tengah (median) dari kolom height

Cell7: Menghitung nilai yang paling sering muncul (modus) dari kolom height

```
df.var(numeric_only=True)
```

	0
Height	268.149162
Weight	1048.633267
Index	1.836168

dtype: float64

```
df.std(numeric_only=True)
```

	0
Height	16.375261
Weight	32.382607
Index	1.355053

dtype: float64

```
Q1 = df['Height'].quantile(0.25)
print("Q1 : ",Q1)

Q3 = df['Height'].quantile(0.75)
print("Q3 : ",Q3)

IQR = Q3 - Q1
print("IQR Height:", IQR)
```

```
Q1 : 156.0
Q3 : 184.0
IQR Height: 28.0
```

Cell8: Menghitung variansi untuk semua kolom numerik pada dataframe

Cell9: Menghitung standar deviasi dari kolom numerik (seberapa besar penyebaran data)

Cell10: Menghitung kuartil pertama (Q1), kuartil ketiga (Q3), dan interquartile range (IQR) kolom height

```
df.describe()
```

	Height	Weight	Index
count	500.000000	500.000000	500.000000
mean	169.944000	106.000000	3.748000
std	16.375261	32.382607	1.355053
min	140.000000	50.000000	0.000000
25%	156.000000	80.000000	3.000000
50%	170.500000	106.000000	4.000000
75%	184.000000	136.000000	5.000000
max	199.000000	160.000000	5.000000

```
correlation_matrix = df.corr(numeric_only=True)

print("Matrix korelasi:")
print(correlation_matrix)
```

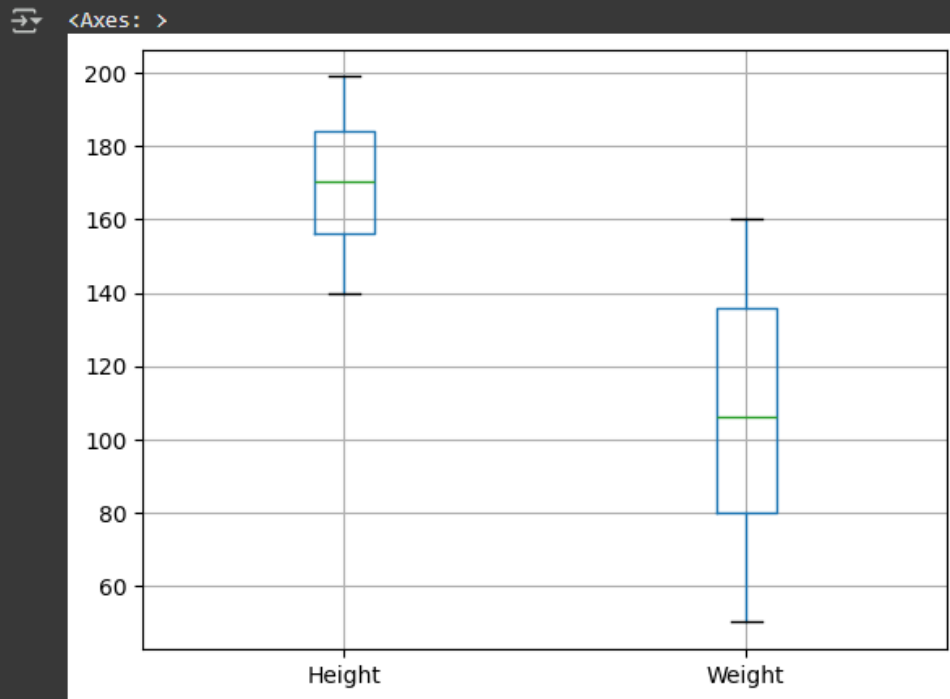
```
Matrix korelasi:
      Height  Weight  Index
Height  1.000000  0.000446 -0.422223
Weight  0.000446  1.000000  0.804569
Index   -0.422223  0.804569  1.000000
```

Cell11: Menampilkan statistik deskriptif otomatis (mean, std, min, max, kuartil) dari semua kolom numerik

Cell12: Menghitung dan menampilkan matriks korelasi antar variabel numerik dalam dataframe

```
import pandas as pd
import numpy as np

df.boxplot(column=['Height', 'Weight'])
```



Membuat boxplot untuk kolom height dan weight agar terlihat sebaran data serta outlier-nya.

```

import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

# Ambil data Height
data_height = df["Height"]

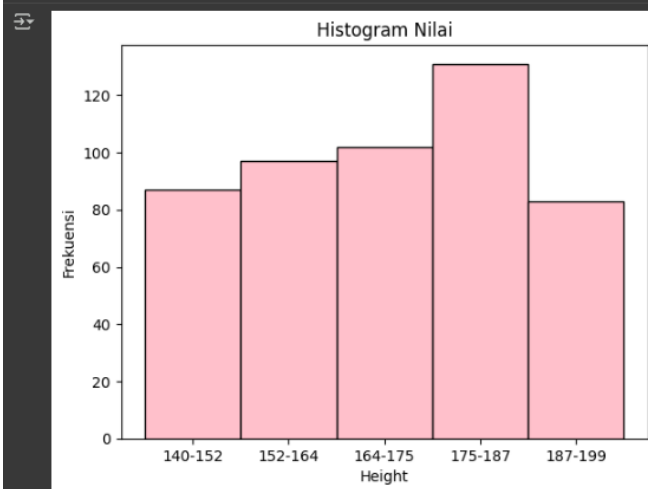
# Buat histogram
n, bins, patches = plt.hist(data_height, bins=5, color='pink', edgecolor='black')

# Tambahkan label
plt.title('Histogram Nilai')
plt.xlabel('Height')
plt.ylabel('Frekuensi')

# Tampilkan rentang frekuensi di sumbu x
bin_centers = 0.5 * (bins[:-1] + bins[1:])
plt.xticks(bin_centers, ['{:0.0f}-{:0.0f}'.format(bins[i], bins[i+1]) for i in range(len(bins)-1)])

# Tampilkan histogram
plt.show()

```



Membuat histogram kolom height untuk melihat distribusi frekuensi tinggi badan dalam lima kelompok.

```

import pandas as pd
import matplotlib.pyplot as plt

# Buat DataFrame contoh
data = {
    'Nilai1': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    'Nilai2': [2, 4, 6, 8, 10, 12, 14, 16, 18, 20]
}

df2 = pd.DataFrame(data)

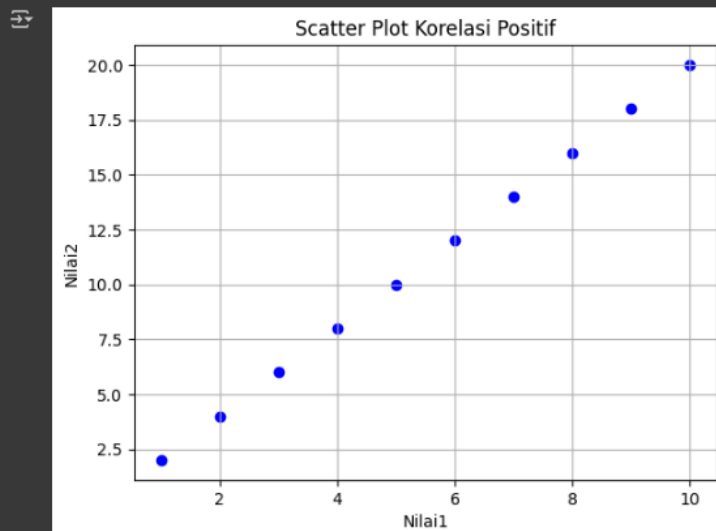
# Buat scatter plot
plt.scatter(df2['Nilai1'], df2['Nilai2'], color='blue', marker='o')

# Tambahkan label
plt.title('Scatter Plot Korelasi Positif')
plt.xlabel('Nilai1')
plt.ylabel('Nilai2')

# Tambahkan grid
plt.grid(True)

# Tampilkan plot
plt.show()

```



Membuat scatter plot yang menunjukkan korelasi positif (nilai1 naik → nilai2 ikut naik).

```

import pandas as pd
import matplotlib.pyplot as plt

# Buat DataFrame contoh
data = {
    'Nilai1': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    'Nilai2': [10, 9, 8, 7, 6, 5, 4, 3, 2, 1]
}

df3 = pd.DataFrame(data)

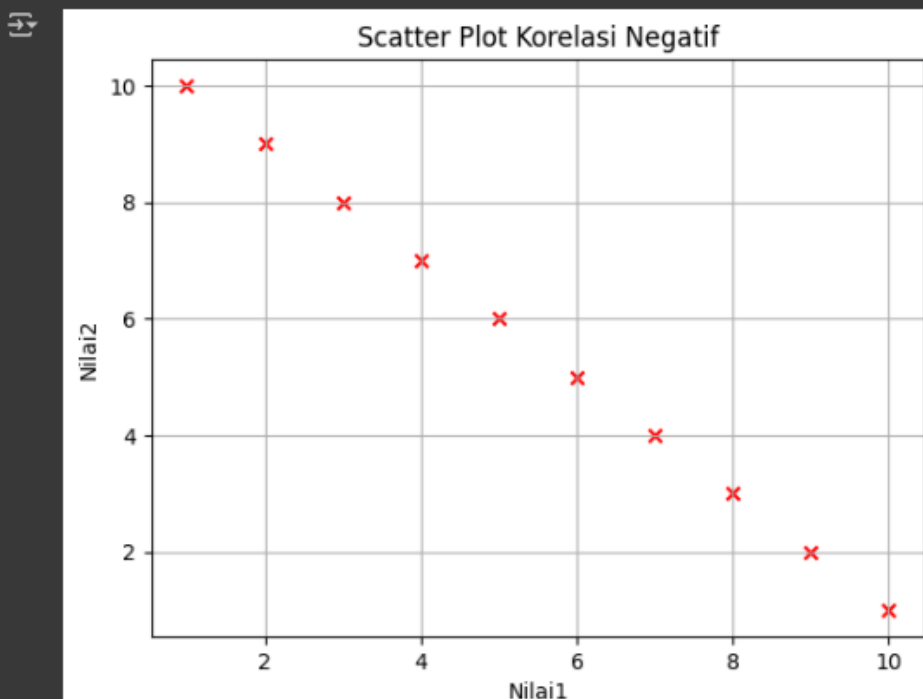
# Buat scatter plot
plt.scatter(df3['Nilai1'], df3['Nilai2'], color='red', marker='x')

# Tambahkan label
plt.title('Scatter Plot Korelasi Negatif')
plt.xlabel('Nilai1')
plt.ylabel('Nilai2')

# Tambahkan grid
plt.grid(True)

# Tampilkan plot
plt.show()

```



Membuat scatter plot dengan korelasi negatif (nilai1 naik → nilai2 turun).

```

# Import library
import pandas as pd
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt

# Baca dataset
df = pd.read_csv(path + '/data/day.csv')

# Bagi dataset
train, test = train_test_split(df, test_size=0.2, random_state=42)
train, val = train_test_split(train, test_size=0.1, random_state=42)

# Tampilkan jumlah data
print(f"Training: {len(train)}")
print(f"Validation: {len(val)}")
print(f"Testing: {len(test)}")

```

```

Training: 525
Validation: 59
Testing: 147

```

Membaca dataset day.csv, lalu membaginya menjadi training (80%), validation (10% dari training), dan testing (20%), kemudian menampilkan jumlah data masing-masing bagian.

```
print("\nTraining:\n", train.head())
print("\nValidation:\n", val.head())
print("\nTesting:\n", test.head())
```

Training:

	instant	dteday	season	yr	mnth	holiday	weekday	workingday	\
657	658	2012-10-19	4	1	10	0	5	1	
163	164	2011-06-13	2	0	6	0	1	1	
305	306	2011-11-02	4	0	11	0	3	1	
111	112	2011-04-22	2	0	4	0	5	1	
538	539	2012-06-22	3	1	6	0	5	1	

	weathersit	temp	atemp	hum	windspeed	casual	registered	\
657	2	0.563333	0.537896	0.815000	0.134954	753	4671	
163	1	0.635000	0.601654	0.494583	0.305350	863	4157	
305	1	0.377500	0.390133	0.718750	0.082092	370	3816	
111	2	0.336667	0.321954	0.729583	0.219521	177	1506	
538	1	0.777500	0.724121	0.573750	0.182842	964	4859	

cnt

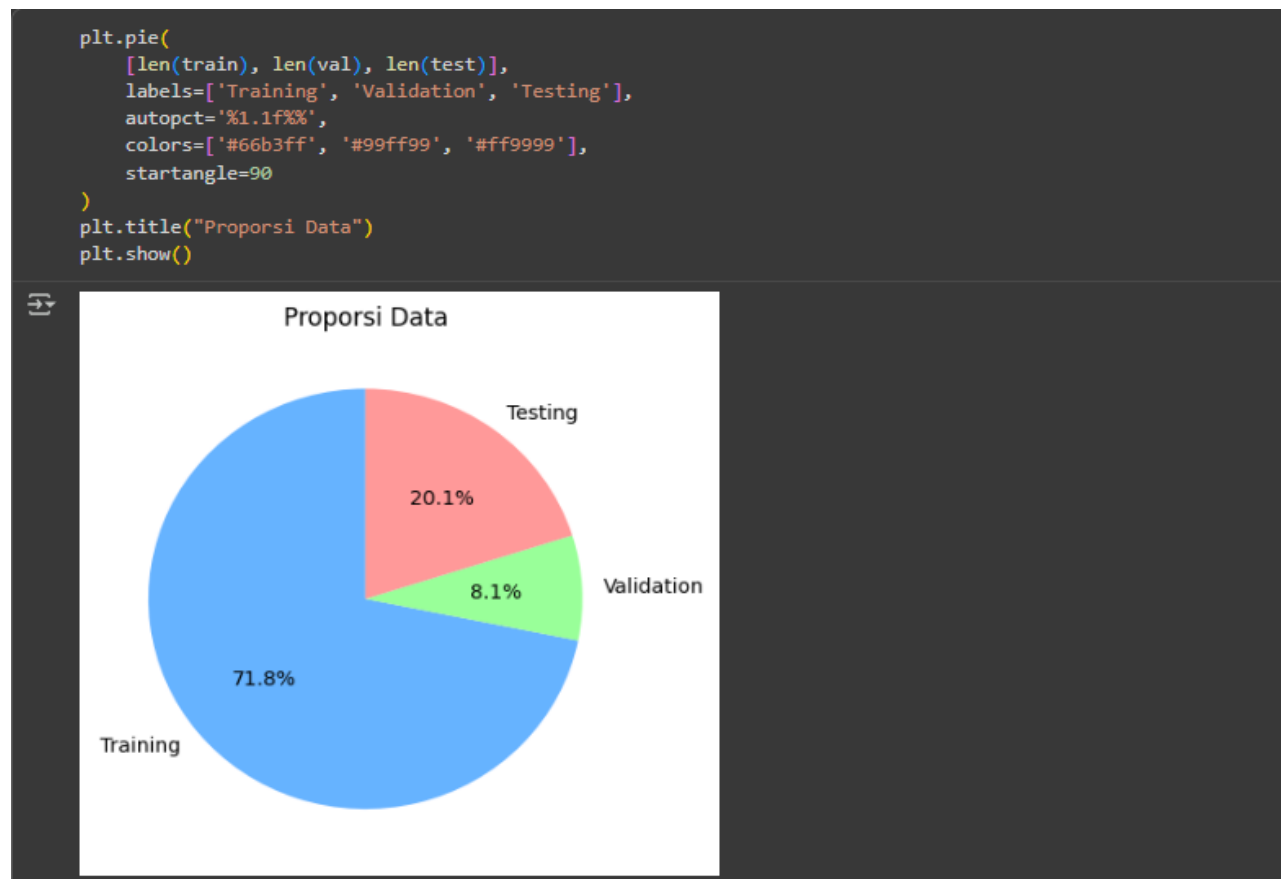
657	5424
163	5020
305	4186
111	1683
538	5823

Validation:

	instant	dteday	season	yr	mnth	holiday	weekday	workingday	\
325	326	2011-11-22	4	0	11	0	2	1	
410	411	2012-02-15	1	1	2	0	3	1	
92	93	2011-04-03	2	0	4	0	0	0	
47	48	2011-02-17	1	0	2	0	4	1	
508	509	2012-05-23	2	1	5	0	3	1	

	weathersit	temp	atemp	hum	windspeed	casual	registered	\
325	3	0.416667	0.421696	0.962500	0.118792	69	1538	
410	1	0.348333	0.351629	0.531250	0.181600	141	4028	
92	1	0.378333	0.378767	0.480000	0.182213	1651	1598	
47	1	0.435833	0.428658	0.505000	0.230104	259	2216	
508	2	0.621667	0.584612	0.774583	0.102000	766	4494	

Menampilkan 5 baris pertama dari masing-masing dataset (training, validation, testing) sebagai bukti hasil pembagian data.



Membuat diagram pie untuk menampilkan proporsi dataset (Training, Validation, Testing)

Link GitHub: <https://github.com/CeperMail/MachineLearning.git>